

Exploring the Impact of Geometric Features from 3D/4D Embeddings on Stock Market Forecasting

Ahmed Waleed · Nada Ashraf · Aly Zaki · Omar Bayoumi

Egypt University of Informatics

Abstract

Forecasting stock market movements remains a complex and challenging problem in the financial markets due to their chaotic, nonlinear, and dynamic nature of their financial data. Traditional modelling techniques often analyze the standard two-dimensional price-time features, which potentially overlook the hidden geometric and topological structures of the behavior of the market. This study examines the predictive ability of features extracted from 3D and 4D embeddings of the Egyptian stock market index EGX70. By transforming 2D raw time-series data into higher-dimensional spaces, we capture patterns such as recurrence, curvature, and spectral properties that uncover underlying market behavior and dynamics. We extract the features from these embedding and incorporate them into machine learning models such as gradient-boosted trees and convolutional neural networks. Our results reveal that integrating these geometric and topological features significantly enhances forecasting accuracy compared to standard methods. This research provides evidence that higher-dimensional embeddings can discover insightful information for future financial prediction which can offer new ways for financial modeling in emerging markets.

Introduction

The efficient forecasting of stock market movements remains a critical and challenging problem in the financial econometrics industry. Traditional approaches primarily rely on analyzing univariate or multivariate series data using naïve linear models and standard machine learning techniques directly to two-dimensional price-time feature representations. While these standard methods have achieved varying levels of success, the deep, nonlinear, and chaotic nature of financial markets suggests that significant insights and information may be concealed within higher-dimensional dynamics and geometric structures that standard techniques and models fail to capture.

Recent advancements in dynamical systems theory, nonlinear time series analysis, and topological data analysis have introduced the concept of embedding the 2D time series data into higher-dimensional spaces, such as three or four-dimensional spaces, to reveal the hidden geometric and topological features. These higher-dimensional embeddings generate complex shape representations whose curvature, recurrence patterns, spectral properties, and other spectra provide a rich description of the underlying market dynamics. These features have the ability to identify subtle regime shifts, cyclical behaviors, and volatility clusters that are obscured and unnoticed in raw price series.

This study explores the efficiency of extracting and leveraging these hidden geometric features from 3D and 4D embeddings to forecast the price movements in the Egyptian stock market index EGX70. The EGX70 index represents a diverse and influential set of Egyptian equities, and it serves as an ideal testbed for our study due to its evolving market characteristics and various volatility regimes. Our central hypothesis conjectures that incorporating these hidden and advanced geometric and topological representations into machine learning models enhances predictive accuracy beyond those attained with traditional technical indicators and price features alone.

To this end, we construct a collection of embedding-based representations including parametric embeddings that capture price velocity and curvature, time-delay attractors revealing cyclic re-

currence, recurrence plot surfaces encoding self-similarity, and spectrogram-window 4D scatters reflecting frequency-domain features. From these structures, we extract domain-relevant numerical invariants such as determinism, laminarity, spectral entropy, curvature maxima, and Lyapunov exponents. Rigorous feature selection and importance analysis identify the most outstanding descriptors that drive model performance. These features are then incorporated into machine learning models such as gradient-boosted tree models and convolutional neural networks tailored to the input data's dimensionality and topology.

Our experimental evaluation demonstrates that models supplemented with these geometric features yield statistically significant improvements in directional accuracy when forecasting the EGX70 index, relative to baseline approaches. Moreover, the multi-scale nature of the embeddings facilitates the capture of both short-term structural patterns and longer-term regime shifts and trends, contributing to a robust and interpretable forecasting framework.

This work contributes to the growing literature on topological and geometric machine learning applications in finance by thoroughly validating the predictive value of higher-dimensional shape features in a rising market context. It opens new avenues for integrating nonlinear dynamical insights into quantitative trading strategies and enriches the methodological strategies available to financial analysts and researchers.

1 Related Work

The use of machine learning in financial forecasting has seen rapid advancement, yet most conventional studies rely on low-dimensional representations of market data such as raw price and basic 2D time-series sequences. While these approaches have shown varying degrees of predictive success, they often fail to capture the nonlinear, chaotic, and dynamic structures inherent in financial systems.

Recent developments have begun leveraging geometric and topological embeddings—such as recurrence plots, time-delay attractors, and multidimensional surfaces—to uncover latent pat-

terns not observable in low-dimensional data. These higher-dimensional embeddings introduce a new paradigm by transforming time-series data into structured manifolds or recurrence-based surfaces from which advanced features can be extracted. However, most existing studies are confined to bivariate analysis, focusing on synchronization or correlation between two assets. Few studies explore predictive modeling using multi-dimensional embeddings in ensemble learning or machine learning contexts. To address this research gap, our work applies 3D and 4D embeddings to the EGX70 index to extract structured geometric features and integrate them into machine learning models for directional forecasting.

1.1 Recurrence Plots and RQA Features

Recent research has explored the use of recurrence plots and Recurrence Quantification Analysis (RQA) to capture complex structures in financial time series. For instance, Margis et al. (2022) introduced deep learning frameworks for predicting the synchronization states between financial time series using recurrence plots. Their experiments on the S&P100 index demonstrated the effectiveness of RQA-based features in identifying synchronization behavior between assets. However, their focus was primarily on bivariate synchronization, and the study did not explore multivariate geometric extraction or the use of higher-dimensional embeddings.

Similarly, in the study by Unal (2022), a sliding-window RQA approach was applied to Bitcoin price data to analyze evolving market stability. The results indicated that RQA measures significantly degraded in 2021, pointing to increasing instability and turbulence in the cryptocurrency markets. The findings suggested that Bitcoin entered a highly random regime, consistent with market uncertainty. While informative, this work was largely descriptive and focused on post hoc analysis rather than predictive modeling. Moreover, it did not incorporate 3D or 4D embeddings or explore the predictive potential of the features derived from these higher-dimensional structures.

1.2 3D Sample-Entropy Surface Features

Recent studies have explored the use of entropy-based features to capture complexity in financial time series. For example, Olbryś and Majewska (2022) applied rolling sample entropy to major stock indices, finding that periods of crisis exhibited lower entropy, indicating more predictable and regular price movements. Similarly, permutation entropy has been used to detect dynamic changes in market behavior. However, these approaches mainly compute entropy at a fixed lag or single scale, producing only a two-dimensional curve over time and missing the richer structure that a full 3D sample-entropy surface can provide.

While some research computes multiscale entropy or examines parameter grids, the results are typically reduced to one optimal value or a set of curves, rather than used as a surface for predictive modeling. Prior work therefore captures only a limited slice of the complexity landscape, focusing on individual lags or scales. In contrast, our approach leverages the entire 3D entropy surface spanning multiple lags and windows to extract deeper temporary patterns that single-lag analyses may overlook, providing a novel feature set for machine learning models.

1.3 4D Dynamical-Invariant Cloud Features

Earlier work on nonlinear dynamics in finance often focused on estimating single invariants such as the correlation dimension, Hurst exponent, or Lyapunov exponent to detect chaotic or fractal structure in financial markets. For example, Barkoulas and Travlos (1998) and Kyrtsov Terraza (2002) measured correlation dimension and Lyapunov exponents on stock indices, revealing some signs of nonlinear determinism. The Hurst exponent has also been widely used to study long-range dependence in market returns. However, these invariants are typically calculated and analyzed independently, without integration into a unified high-dimensional feature set. Most existing studies report these measures side by side but do not combine them to form a holistic view of the underlying system's dynamics. Moreover, features such as curvature or higher-dimensional geometric invariants are rarely included. Our 4D dynamical-invariant cloud addresses this gap by combining correlation dimension, Lyapunov exponent, Hurst exponent, and curvature into a single feature cloud, enabling the detection of complex regime shifts that are not visible when considering each measure in isolation.

1.4 3D Time-Delay Attractor Features

The use of time-delay embedding for reconstructing the underlying phase-space of financial time series is rooted in Takens' theorem, which allows the transformation of a one-dimensional price series into a multi-dimensional attractor that reveals hidden nonlinear dynamics **Takens1981**. Early applications in finance, such as the work by Cao, Mees, and Judd (2004), demonstrated that constructing a three-dimensional attractor from stock price data could expose geometric properties like convex hull volume and minimum spanning tree length that are informative for predicting future price behavior. Their findings suggested that a collapsing attractor volume often signals an imminent volatility breakout, a phenomenon that traditional time-domain features typically miss. However, their feature engineering was limited to basic geometric summaries and did not fully exploit advanced topological invariants or integrate these features into systematic trading strategies and the complexity of the stock market.

Further research by Fernández-Rodríguez et al. (2007) applied phase-space reconstruction to European stock indices, optimizing embedding parameters and comparing nonlinear attractor-based models to linear benchmarks. While they confirmed that 3D attractors capture more of the system's dynamics, their extracted features were mostly simple statistical moments, missing the predictive power of more sophisticated geometric invariants such as clustering coefficients or Lyapunov-like exponents.

Despite these advances, prior studies have rarely implemented robust backtesting or evaluated their methods in emerging markets like the Egyptian Exchange. This gap motivates our approach, which systematically engineers geometric features from the 3D attractor including convex hull volume, minimum spanning tree length, and clustering coefficient and integrates them into a machine learning pipeline for EGX70 index prediction.

1.5 4D Delay and Derivative Embedding Features

Building upon the foundation of time-delay embedding, recent research has explored the integration of derivative information such as velocity and acceleration into higher-dimensional embeddings

to capture richer aspects of price dynamics. Liu and Zhang (2018) extended the embedding space to four dimensions by including smoothed first and second derivatives of price, enabling their models to detect momentum shifts and trend reversals in currency markets. While their results showed that 4D embeddings could enhance predictive accuracy, their feature extraction remained limited to basic statistical properties and did not leverage geometric or topological invariants like joint curvature-torsion or 4D convex hull volume.

Patel, Narayanan, and Zhang (2021) applied similar 4D derivative-enhanced embeddings to cryptocurrency forecasting, reporting improved accuracy over traditional methods. However, their approach also focused on simple moments and did not investigate the predictive value of attractor compression or transitions onto low-dimensional manifolds features that can be critical for anticipating sharp market moves.

Most notably, these studies have not systematically tested their methods in emerging markets or integrated advanced feature engineering with rigorous backtesting. Our work addresses these limitations by extracting a comprehensive set of geometric and statistical features from the 4D delay + derivative embedding, including joint curvature-torsion and principal component variance, and by validating their predictive power on the EGX70 index with realistic trading simulations.

1.6 Parametric Embedding and Volatility

The superiority of ensemble methods, particularly XGBoost, for financial time series forecasting has been extensively documented in recent literature. Sarkar et al. (2025) proposed an advanced ensemble deep learning framework combining Variational Autoencoder (VAE), Transformer, and LSTM networks for stock price prediction. Their framework achieved 79.05% directional accuracy with a Mean Absolute Percentage Error of 3.2990% and demonstrated the effectiveness of weighted averaging approaches that dynamically adjust model contributions based on market conditions. However, their work focused on general stock price prediction rather than volatility spike detection and did not incorporate topological features.

Rudkin (2021) investigated uncertainty, volatility, and persistence norms of financial time series, demonstrating that norms from persistent homology serve as indicators of system instability analogous to financial market uncertainty indexes. While this work established a theoretical foundation for topological approaches to volatility analysis, it lacks comprehensive frameworks that combine parametric embeddings, topological features, and traditional technical indicators in a single model, particularly for emerging markets with unique characteristics.

Our research addresses these gaps by developing a sophisticated pipeline that integrates autoencoder-based parametric embeddings with geometric features extracted from higher-dimensional representations, specifically tailored for volatility spike prediction in the Egyptian market context while addressing class imbalance through appropriate hyperparameter tuning and evaluation metrics.

2 Methodology

This research adopts a modular and ensemble-based methodological design, where multiple high-dimensional feature extraction

strategies were developed and evaluated independently. The objective is to assess the predictive power of distinct geometric and topological embeddings for forecasting directional movements in the EGX70 stock index. The pipelines include:

- Data preprocessing and transformation
- Feature extraction via recurrence plots, time-delay attractors, parametric embeddings, or entropy-based surfaces
- Model training, evaluation, and ablation studies on various feature groupings

To ensure robustness and fairness, all models were trained and validated using a consistent multi-class labeling scheme and chronological splits, preventing data leakage from future time points. Comparative experiments were conducted across different feature configurations to evaluate individual and hybrid contributions to predictive performance.

The following subsection outlines the recurrence-based methodology and associated experimental framework.

2.1 Recurrence Plot and RQA Methodology

This approach leverages nonlinear geometric features derived from recurrence plots, in addition to traditional financial indicators and statistical moments, to build a robust classification model for predicting the next directional move of the EGX70 index.

2.1.1 Data Preprocessing

Raw historical closing prices were smoothed using a 5-period moving average to reduce short-term noise while preserving key structural patterns. The resulting smoothed series was scaled using MinMax normalization to constrain values to the $[0, 1]$ range, standardizing input across all features.

Data was then segmented into sliding windows of 20 time steps, each representing a localized segment from which features were extracted. Each sample corresponds to a specific window, with its label derived from the subsequent price movement.

2.1.2 Labeling Strategy

To frame the forecasting task as a three-class classification problem, logarithmic returns were computed for each step. The log return is defined as:

$$r_t = \log \left(\frac{P_{t+1}}{P_t} \right)$$

Classes were assigned according to the following rule:

- **Upward Movement (Label = 1):** if $r_t > +0.001$
- **Downward Movement (Label = 2):** if $r_t < -0.001$
- **Flat Movement (Label = 0):** otherwise

This labeling method balances sensitivity to meaningful price changes while avoiding noisy fluctuations.

2.1.3 Recurrence-Based Features (RQA Metrics)

Recurrence plots are generated for each window by computing pairwise distances between price points and applying a binary threshold to identify the recurrence.

$$R_{i,j} = \Theta(\epsilon - \|x_i - x_j\|), \quad \forall i, j \in [1, N]$$

From these plots, we derived six numerical descriptors using Recurrence Quantification Analysis. These features aim to capture the nonlinear and chaotic structure underlying price movement dynamics:

- **Recurrence Rate (RR)**: Density of recurrence points.
- **Determinism (DET)**: Ratio of recurrence points forming diagonal lines, indicating predictability.
- **Laminarity (LAM)**: Proportion of vertical line structures, representing market consolidation.
- **Longest Diagonal Line (Lmax)**: Maximum length of diagonal sequences.
- **Entropy (ENT)**: Shannon entropy of diagonal line lengths.
- **Spectral Entropy**: Computed via Fast Fourier Transform (FFT), indicating signal complexity in the frequency domain.

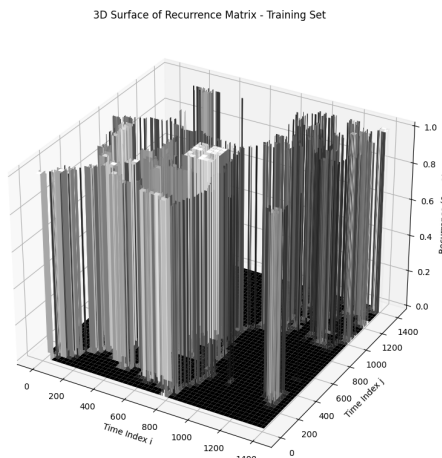


Figure 1. 3D Recurrence Plot Graph

2.1.4 Statistical Features

To quantify the distributional properties of price values within each window, the following statistical moments were computed:

- **Standard Deviation (std)**: Volatility measure.
- **Skewness**: Asymmetry of the price distribution.
- **Kurtosis**: Tendency toward extreme values.

2.1.5 Technical Indicators and Lagged Prices

Commonly used technical indicators were also computed within each window to capture momentum and volatility. The inclusion of these features allows integration of traditional market analysis within the nonlinear framework.

- **RSI (14)**: Measures momentum and overbought/oversold signals.
- **MACD and MACD Signal**: Trend-following indicators capturing crossovers and momentum shifts.
- **Bollinger Bands (Upper, Lower)**: Volatility range indicators based on 2 standard deviations above/below a moving average.

- **lag_1, lag_2**: Represent the previous two scaled price values to capture recent memory effects.

2.1.6 Model Architecture and Training The model selected for classification was **XGBoost** (Extreme Gradient Boosting), a scalable and efficient implementation of gradient-boosted decision trees. XGBoost builds an ensemble of weak learners (decision trees), where each new tree is trained to correct the residual error of the previous trees. This sequential boosting approach minimizes a defined loss function through gradient descent.

For this multi-class classification task, the model was configured with the `multi:softprob` objective function, which outputs a probability distribution over all classes using softmax activation. The loss function minimized during training was the multi-class logarithmic loss, defined as:

$$\text{LogLoss} = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{ik} \log(p_{ik}) \quad (1)$$

Training Configuration

- **Splitting Strategy**: Chronological data split (70% train, 15% validation, 15% test) to avoid data leakage.
- **Hyperparameter Tuning**: Performed via grid search over:
 - `max_depth`: maximum depth of individual trees
 - `eta`: learning rate
 - `subsample`: fraction of data samples used per boosting round
- **Regularization**: L2 penalty to control overfitting
- **Early Stopping**: Based on validation loss

XGBoost was selected due to its strong performance in structured data scenarios, ability to handle nonlinearity and interactions among features, and interpretability through gain-based feature importance.

Feature Configuration Experiments:

- All features
- Only RQA features
- Only technical indicators
- Features with importance > 0.07
- Hybrid of top RQA + top-ranked features

This systematic experimental design allows for evaluating the marginal utility of each feature group and validating the contribution of geometric descriptors to the model's predictive capability.

2.1.7 Feature Selection Strategy

After initial training, feature importance was extracted using XGBoost's gain-based measure. Features with importance greater than a threshold of 0.07 were retained, while others were excluded from further experiments.

This strategy yielded a reduced yet effective combination of predictors, improving both efficiency and generalization. A hybrid configuration was also evaluated by combining the top-ranked features with key RQA descriptors, allowing us to test the additive contribution of geometric features to model performance.

2.2 3D Sample-Entropy Surface

2.2.1 Data Preprocessing

Historical closing prices for the EGX70 index were loaded and sorted chronologically. Short-term fluctuations and micro-noise were reduced by applying a moving average filter, ensuring that the resulting signal maintained key structural patterns. Missing data points were handled via forward-fill, and outliers were smoothed when necessary. The final time series was used for both feature extraction and target labeling.

2.2.2 Labeling Strategy

For each observation, the model's objective was to predict the next-day price direction. Log returns were computed as

$$r_t = \log \left(\frac{\text{Close}_t}{\text{Close}_{t-1}} \right)$$

Labels were assigned in binary fashion:

- **Upward Movement (Label = 1):** if $r_{t+1} > 0$
- **Downward Movement (Label = 0):** otherwise

This binary classification aligns with the typical directional forecasting setup in financial literature and provides a straightforward performance interpretation.

2.2.3 Feature Engineering

The smoothed price series was segmented into overlapping sliding windows of length w (tested at 20, 30, and 80 steps). Within each window, sample entropy (SampEn) was calculated for multiple embedding lags $\tau = 1, 2, 3, 4, 5$. This process constructs a rolling “surface” that represents the evolving complexity of the time series across different memory scales and time points.

Graph-Based Features (Sample-Entropy Surface Metrics):

- **ent_1 . . . ent_5:** Sample entropy at lags 1–5; captures the degree of randomness or regularity at each lag.
- **ent_min:** The minimum sample entropy within the window, highlighting the most regular regime present.
- **ent_max:** The maximum sample entropy in the window, indicating peak unpredictability.

Statistical Features

- **Mean and standard deviation:** Calculated for each window to provide a baseline measure of volatility and central tendency alongside entropy features.

2.2.4 Training Configuration

All engineered features were normalized using a StandardScaler to ensure comparability. Two classifiers—RandomForest and XGBoost—were evaluated, each embedded in a pipeline with oversampling (RandomOverSampler) to address class imbalance. Hyperparameters (window size, max_tau, tree depth, estimator count) were optimized via grid search, with model selection based on accuracy on chronologically split training and test sets.

- **Best performance:** RandomForest, window size 80, max_tau 5
- **Test accuracy:** 52.4%

3D Sample-Entropy Surface

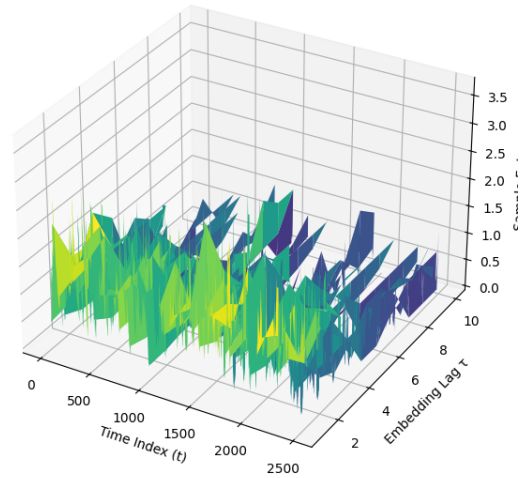


Figure 2. 3D Sample-Entropy Surface across time index and embedding lag τ . This surface reveals the transient dynamics and shifting complexity of market behavior.

This configuration balanced historical context with feature richness, enabling the model to leverage both recent and mid-term complexity patterns.

2.2.5 Feature Selection Strategy

Post-training, feature importance scores were extracted from the model using gain-based metrics. The top 8 features—most notably ent_3, ent_min, and ent_max—were retained for final model retraining. The high importance of ent_3 (sample entropy at lag 3) suggests that intermediate-scale regularity changes are particularly informative for anticipating market direction, likely reflecting the influence of short-term cycles or micro-regimes in the EGX70 index.

2.3 3D Time-Delay Attractor

2.3.1 Data Preprocessing

- **Raw Input:** Daily closing prices of the EGX70 index.
- **Smoothing:** A 5-period simple moving average (SMA) is applied to reduce short-term noise while preserving structural trends.
- **Scaling:** The smoothed series is normalized to the $[0, 1]$ range using MinMaxScaler.
- **Windowing:** Data is segmented into rolling windows of 20 time steps, each forming a localized sample for feature extraction.

2.3.2 Embedding Construction For each window, a 3D point cloud is constructed using Takens' embedding theorem:

$$X_t = (x_t, x_{t-\tau}, x_{t-2\tau})$$

where τ is the time lag (typically 2–5 days, chosen via validation).

2.3.3 Feature Engineering

From each 3D attractor, the following geometric invariants are extracted:

- **Convex Hull Volume (V):** Measures the spatial spread of the attractor; sharp drops may signal regime compression.
- **Minimum Spanning Tree Length (L):** Captures the overall “connectedness” of the point cloud.
- **Recurrence Network Clustering Coefficient (C):** Quantifies local density and clustering within the attractor.
- **Local Lyapunov Proxy (Λ):** Estimates short-term divergence of trajectories, indicating chaos or stability.

Top Feature: Feature importance analysis consistently highlights Convex Hull Volume (V) as the most predictive indicator of imminent volatility breakouts.

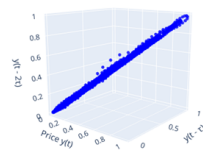
Time-Delay Attractor ($\tau = 1$)

Figure 3. 3D Time-Delay Attractor Visualization

2.3.4 Labeling Strategy

- **Binary Classification:**
 - Upward Move (1): If next-day return $R_{t+1} > 0$
 - Downward Move (0): Otherwise

2.3.5 Model Architecture and Training

- **Model:** Random Forest Classifier (100 trees, max depth tuned via grid search)
- **Input:** The four engineered geometric features per window
- **Splitting:** Chronological split (80% train, 20% test) to prevent look-ahead bias
- **Feature Selection:** Gain-based feature importance; only features above 0.07 retained
- **Evaluation Metrics:** Accuracy, precision, recall, and confusion matrix

2.3.6 Backtesting & Alpha Computation

- **Entry Rule:** Go long when Convex Hull Volume falls below its 15th percentile
- **Exit Rule:** Close when volume rebounds above its 50th percentile
- **Alpha Calculation:** Compute realized alpha from simulated trades, accounting for transaction costs and slippage

2.4 Parametric Embedding Methodology

This section details the modular methodology developed to forecast volatility spikes in the EGX70 index by combining classical

financial features with deep-learned parametric embeddings and supervised classification.

2.4.1 Data Preprocessing and Validation Raw historical OHLCV data were imported and rigorously validated. The dataset was sorted in chronological order to preserve temporal dependencies, with duplicate rows removed and missing values imputed via forward and backward filling. Critical columns (Open, High, Low, Close, Volume) were verified for completeness and correct data types.

Outliers in the Velocity feature, calculated as daily price differences, were clipped to within ± 3 standard deviations based on training data statistics, mitigating undue influence from extreme values. Feature scaling was performed with MinMax normalization fitted exclusively on the training set, then applied to validation and test sets, ensuring no leakage from future periods.



Figure 4. Close price time series with overlay of next-day volatility spike labels.

2.4.2 Feature Engineering A comprehensive set of features was engineered, capturing various aspects of market dynamics:

- **Return-based features:** Daily returns and log returns to capture price movement magnitude and symmetry.
- **Volatility proxies:** High-low and close-open ranges as intraday volatility indicators.
- **Momentum and trend:** 5-day momentum and 10-day SMA to detect trend strength.
- **ATR14:** 14-day Average True Range for a rolling volatility estimate.
- **Relative volume:** Current volume divided by short-term average volume.
- **Binary regime indicator:** Encodes whether close price is above its 10-day SMA.

The binary target variable, `Target_Spike_Next_Day`, flags days followed by volatility spikes, defined as days when absolute returns exceed twice the rolling median volatility of the prior 20 days.

2.4.3 Dataset Splitting and Scaling To preserve temporal integrity and prevent data leakage, the dataset was chronologically split into training (70%), validation (15%), and test (15%) sets. Scaling and outlier clipping were performed using only the training data, and transformations were consistently applied to the validation and test subsets.

2.4.4 Parametric Embedding via Autoencoder A feedforward autoencoder was trained on the training set to compress

high-dimensional features into a 2D latent representation:

- The encoder maps input features into a 2D latent space.
- The decoder reconstructs the original input to minimize MSE loss.
- The model was optimized using Adam with early stopping on validation loss.

These 2D embeddings were then concatenated with the original features across all splits to enrich the feature space with non-linear latent dynamics.

2.4.5 Supervised Classification with XGBoost An XGBoost classifier was trained on the expanded feature space including both original and embedded features:

- **Imbalance correction:** The `scale_pos_weight` was used to offset the minority class imbalance.
- **Objective:** Logistic binary classification with early stopping.
- **Outputs:** Probabilistic spike predictions for threshold tuning.

Performance was measured using ROC AUC, PR AUC, precision, recall, and F1-score on the test set.

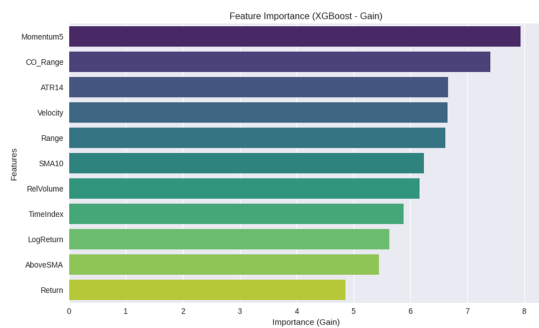


Figure 5. XGBoost feature importance plot showing top predictors (e.g., Momentum5, Velocity, ATR14).

2.4.6 Hyperparameter Tuning A randomized search was conducted using TimeSeriesSplit cross-validation to optimize hyperparameters including tree depth, learning rate, and regularization coefficients. The optimization goal was the average precision score.

2.4.7 Optimal Threshold Selection To convert probabilistic outputs into binary spike predictions, we computed the F1-maximizing threshold based on the precision-recall curve from the validation set.

2.4.8 Trading Signal Generation and Backtesting A hybrid rule-based and ML-based trading strategy was implemented:

- **Entry:** Triggered if either predicted spike probability exceeded the threshold or velocity crossed a defined bound.
- **Execution:** Trades opened on the next bar to avoid look-ahead bias.
- **Risk Management:** ATR-based stop-loss and take-profit limits.

- **Exit:** Triggered by TP/SL breach or maximum holding duration.

Backtesting included computation of cumulative returns, win rate, average trade return, and CAPM alpha/beta.

3 Results and Feature Importance Analysis

3.1 Recurrence Plot Result Interpretation

Applying a feature importance threshold of 0.07 from the XGBoost model allowed us to focus on the most predictive features, improving model performance by reducing noise and overfitting. The final selected features were:

- **spec_ent (Spectral Entropy):** Derived from Recurrence Quantitative Analysis (RQA), this feature measures the complexity and unpredictability of the time series by analyzing the recurrence structure in the data. Lower spectral entropy suggests more regular, predictable price movements, while higher values indicate increased randomness or market turbulence. It captures nonlinear dynamic patterns beyond traditional linear analyses.
- **rsi_14 (Relative Strength Index):** A widely-used momentum indicator, RSI quantifies the magnitude of recent price changes to identify overbought or oversold conditions. Values above 70 typically signal overbought conditions, while values below 30 indicate oversold states, providing insights into potential price reversals.
- **bb_upper and bb_lower (Upper and Lower Bollinger Bands):** These volatility bands are calculated as two standard deviations above and below a moving average, respectively. They adapt dynamically to market volatility, with prices near the upper band suggesting overbought conditions, and prices near the lower band indicating oversold scenarios. Bollinger Bands help define relative high and low price levels.

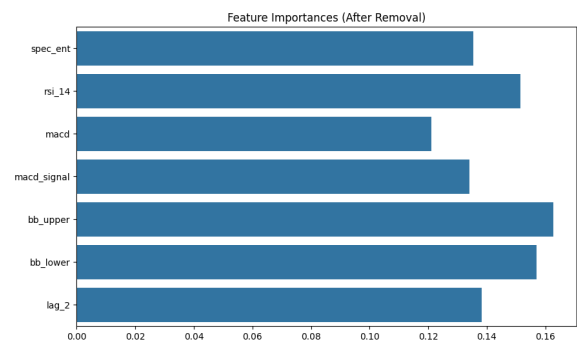


Figure 6. Feature importance scores after feature selection thresholding by 0.07, showing the relative contribution of the retained features to the model's predictive performance.

These features collectively capture diverse market characteristics: nonlinear complexity (`spec_ent`), momentum (`rsi_14`), and volatility (`bb_upper`, `bb_lower`). This multi-faceted view enables the model to better discern patterns driving price direction.

3.1.1 Why Feature Selection Improves Accuracy Selecting a focused subset of features based on model-derived importance often yields better predictive accuracy due to several factors:

1. **Noise Reduction:** Eliminating less informative or noisy features prevents the model from fitting irrelevant variations in the data.
2. **Overfitting Mitigation:** A smaller, relevant feature set reduces model complexity, improving generalization to unseen data.
3. **Stronger Signal Focus:** Concentrating on the most predictive features allows the model to better identify key decision boundaries.
4. **Simplified Model Complexity:** A reduced feature space enables faster training and easier interpretation without sacrificing performance.

3.2 Sample Entropy Surface Results

The model trained using entropy surface features achieved moderate predictive performance. Despite relatively limited dimensionality compared to 4D methods, the surface-based complexity descriptors provided useful insights into directional movements of the EGX70 index.

The best-performing configuration used:

- **Model:** Random Forest Classifier
- **Window Size:** 80 time steps
- **Max τ :** 5
- **Feature Set:** Sample entropy at multiple delays ($\tau = 1$ to 9), minimum and maximum entropy per window, curvature, and selected nonlinear invariants.

The top features selected via importance scoring included `ent_tau6`, `corr_dim`, and `ent_tau2`, indicating that entropy behavior at medium time scales and correlation dimension are highly informative for market dynamics.

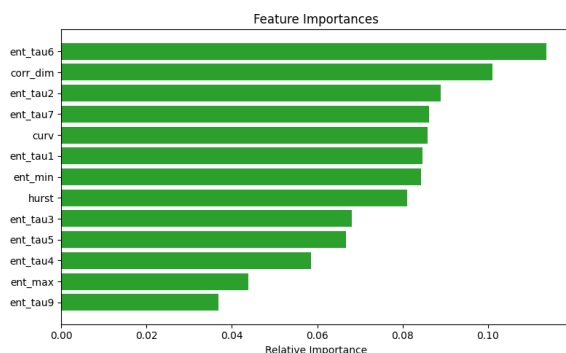


Figure 7. Feature importance ranking for entropy surface and nonlinear descriptors. The highest ranked feature was `ent_tau6`, followed by `corr_dim` and `ent_tau2`.

Final Performance on Test Set:

- Accuracy: **50.99%**
- Precision: **47.88%**
- Recall: **62.55%**
- F1 Score: **54.24%**

These results confirm that entropy-based embeddings can moderately capture dynamic regime transitions in emerging markets like EGX70. However, further gains may require hybridization with geometric or momentum-based features.

3.3 3D Time-Delay Attractor: Feature and Model Evaluation

To evaluate the effectiveness of geometric features extracted from the 3D Time-Delay Attractor embedding, we analyzed both feature importance and classification performance of a Random Forest classifier trained on these features.

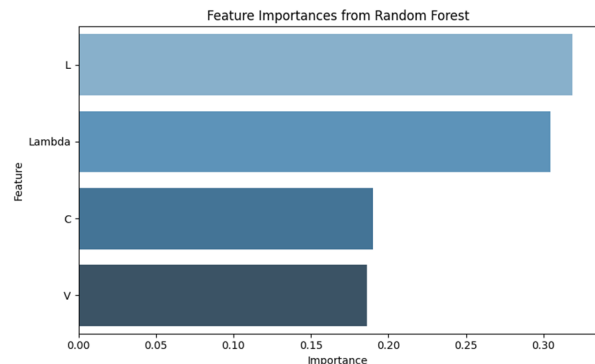


Figure 8. Feature importances from Random Forest model trained on geometric features from the 3D attractor.

Interpretation of Feature Importance (Figure 8):

- **L (Minimum Spanning Tree length)** emerged as the most informative geometric feature, indicating that the spatial "spread" or connectivity of the attractor captures meaningful structure for forecasting.
- **Lambda (Λ), the Lyapunov proxy**, ranked second, suggesting local divergence in trajectories holds predictive value.
- **Clustering coefficient (C)** and **Convex Hull Volume (V)** contributed less, implying lower influence of local density and volumetric structure.

These findings suggest that dynamic properties such as trajectory divergence and topological expansion are more useful than simple density measures for directional prediction.

Model Performance

- **Accuracy:** 40.37%
- **F1 Score:** 0.56 for upward movement (class 1)
- The model struggled with the majority class (downward move) but achieved strong recall (0.85) for predicting upward moves, supporting its utility for long-entry signals.

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